Measuring Fairness through Calibration in Insurance Scoring Algorithms

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Motivation: Using Probabilistic Classifiers

- Several applications of (binary) classification require **probabilistic classifiers**.
  - Rather than predicting labels in $\{0, 1\}$, they output the estimated likelihood of the positive class given the features, *i.e.*, **confidence scores** $\in [0, 1]$.
- For example, in **insurance pricing**:
  - Estimation of **mortality scores**, $\mathbb{P}(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$, where $Y \in \{0, 1\}$ indicates the occurrence of death (Sauce et al., 2023).
  - The **probability that an insured has an accident** within the year is estimated and directly used to calculate the pure premium.
- Using a **GLM**, we can estimate

$$\mathbb{E}[Y \mid \boldsymbol{X}] = \mathbb{P}(Y = 1 \mid \boldsymbol{X}) = \overbrace{\frac{\exp[\boldsymbol{X}^\top \boldsymbol{\beta}]}{1 + \exp[\boldsymbol{X}^\top \boldsymbol{\beta}]}}^{\text{“probabilistic scores”}} = s(\boldsymbol{X})$$

Motivation: Machine Learning for Insurance Pricing

- Machine learning **models** are increasingly used by insurers to provide more **personalized premiums** (NAAC, 2022) or to handle **high-dimensional data** (Gao et al., 2018).
- Using proper loss functions, such as the log-loss, we can decompose the divergence d_ϕ between the true probability $\mathbb{P}(Y \mid \boldsymbol{X})$ and the estimated scores $s(\boldsymbol{X})$ (Gneiting and Raftery, 2007; Kull and Flach, 2015)

$$\mathbb{E}\left[d_\phi(s(\boldsymbol{X}), \mathbb{P}_{Y|\boldsymbol{X}})\right] = \textbf{Calibration Loss} + \text{Grouping Loss}.$$

  - To accurately interpret scores as probabilities of occurrence, we need **Calibration Loss** $= 0$.
- "*Guo et al. (2017) have shown that* **modern neural networks** *(NN) are* **poorly calibrated** *and over-confident despite having better performance*."

This Talk

In brief,

1. Using a machine learning binary classifier, we estimate **confidence scores as probabilities of occurrence** of a binary event.
2. The confidence scores do not necessarily have the same **characteristics across groups** defined by a sensitive attribute.

   How can we correct the <u>unfairness</u> in the confidence scores of a binary classifier?

3. In the literature,
   - **Calibration metrics** calculated within sensitive groups,
   - **Mitigation techniques** to achieve fairness with respect to those calibration metrics.

Table of Contents

What the remainder of the talk is about:

1. Presentation of the notion of **calibration** to assess the **confidence scores** of a binary classifier.
2. Review of methods to **measure and visualize calibration** for a binary classifier.
3. Review of **metrics to evaluate unfairness** based on calibration errors across sensitive groups.
4. Application of two **unfairness mitigation techniques** based on calibration correction to a life insurance dataset.

Framework and Notations

- $A \in \mathcal{A} = \{0, 1\}$ denotes a "protected attribute" (*e.g.*, sex or race),
- $\boldsymbol{X} \in \mathcal{X} = \mathbb{R}^d$ are considered "non-protected" explanatory variables,
- $Y \in \mathcal{Y} = \{0, 1\}$ is the binary outcome variable we aim to predict,
- $s : \mathcal{Z} \to [0, 1]$ is a **binary probabilistic classifier**, where $\mathcal{Z} = \mathcal{X} \times \mathcal{A}$,

$$s(\boldsymbol{X}, A) := \widehat{\mathbb{E}}[Y \mid \boldsymbol{X}, A] = \widehat{\mathbb{P}}(Y = 1 \mid \boldsymbol{X}, A).$$

  - We denote the **Bayes-optimal score**

$$\mu(\boldsymbol{X}, A) := \mathbb{E}[Y \mid \boldsymbol{X}, A] = \mathbb{P}(Y = 1 \mid \boldsymbol{X}, A).$$

Framework: Calibration can be assessed for all observations in a dataset as a **measure of performance** for a binary classifier or between groups as a **measure of unfairness**, *e.g.*, for $A = 0$ and $A = 1$.

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Insurance Pricing Criteria: "Overall" Calibration

Global balance:

$$\mathbb{E}\big[Y - s(\boldsymbol{Z})\big] = \mathbb{E}\big[\mu(\boldsymbol{Z}) - s(\boldsymbol{Z})\big] = 0.$$

- Economically, the **portfolio** is self-financing (for Y).

Calibration (Dawid, 1982) (or auto-calibration (Denuit et al., 2021)) "*among insureds to whom a calibrated mortality-risk classifier assigns a probability of 0.6, 60% will actually die within the year*"

$$\mathbb{E}\big[Y - s(\boldsymbol{Z}) \mid s(\boldsymbol{Z})\big] \overset{\text{a.s.}}{=} \mathbb{E}\big[\mu(\boldsymbol{Z}) - s(\boldsymbol{Z}) \mid s(\boldsymbol{Z})\big] \overset{\text{a.s.}}{=} 0.$$

or equivalently, $\mathbb{E}\big[Y \mid s(\boldsymbol{Z}) = p\big] = p$, for all $p \in \mathrm{Supp}\,(s(\boldsymbol{Z}))$.

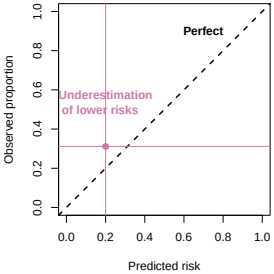
- Economically, **price-based subgroups** $s(\boldsymbol{Z})$ are self-financing (for Y).

Visualizing Calibration

To evaluate the **calibration** of a model s, we can plot its **calibration curve**:

$$\forall p \in [0,1], \quad g : \begin{cases} [0,1] \to [0,1] \\ p \mapsto g(p) := \mathbb{E}[Y \mid s(\boldsymbol{Z}) = p] \end{cases} .$$

- For a confidence score of 0.2, 25 out of 100 in this group are found to have the disease: the risk is **underestimated**.



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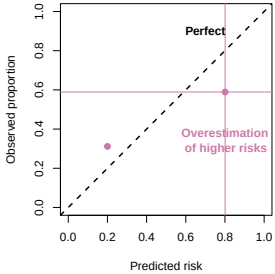
- For a confidence score of 0.8, 60 out of 100 in this group are found to have the disease: the risk is **overestimated**.



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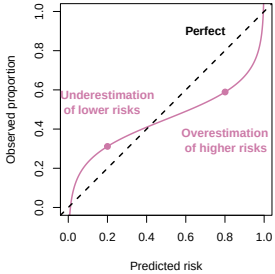

Assessing Calibration

Based on the **distance between the calibration curve g and the bisector**, the **calibration error** of a probabilistic classifier s can be estimated using

- The **Expected Calibration Error** (ECE) (Pakdaman Naeini et al., 2015): using a binning scheme based on score quantiles;
- **Isotonic regression**, which provides an optimal binning scheme on s (Dimitriadis et al., 2021);
- The **Integrated Calibration Index** (ICI) (Austin and Steyerberg, 2019): estimating the calibration curve via local regression (Fernandes Machado et al., 2024).

Framework: When those calibration metrics are calculated over the entire dataset, they serve as a **measure of performance** for a probabilistic binary classifier.

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Overview of Group-Fairness Criteria

- The notion of fairness for groups based on A (*e.g.*, **male** *vs* **female**) differs from *actuarial fairness*, "*requiring premiums to be set proportional to an individual's risk*." (Baumann and Loi, 2023; Arrow, 1963)
- We can use different **group fairness metrics** to evaluate the unfairness of a model s w.r.t. A (Barocas et al., 2023; Charpentier, 2024).
  1. The **Independence** criterion requiring $s(\boldsymbol{Z}) \perp A$, e.g., Demographic Parity (Fernandes Machado et al., 2025);
  2. The **Separation** criterion requiring $s(\boldsymbol{Z}) \perp A \mid Y$, e.g., Equalized Odds;
  3. The **Sufficiency** criterion requiring $Y \perp A \mid s(\boldsymbol{Z})$, e.g., **similar calibration across sensitive groups**.
- Due to **impossibility theorems**, these criteria cannot be mathematically achieved simultaneously (Kleinberg et al., 2016), unless very strict assumptions are met.

Sufficiency Criterion in Insurance Pricing

Why use the **Sufficiency Criterion** (similar calibration errors across sensitive groups) *rather than Independence or Separation to evaluate unfairness in an insurance context?*

- For insurance pricing, compared to the traditional predictive modeling framework, **the true risk of policyholders cannot be measured** at the individual level, which may render some fairness criteria irrelevant (Baumann and Loi, 2023).
- The Sufficiency criterion is widely used in **healthcare** (Obermeyer et al., 2019; Liu et al., 2025) and should also be the case for **life insurance pricing** models.
- In practice, the **cost of fairness** (deviating from *actuarial fairness*) required to achieve Sufficiency is lower than that for Independence or Separation (Liu et al., 2019).

Calibration-based Unfairness Metrics

- **Calibration parity** (Chouldechova, 2017):

$$\forall p \in [0,1],\ \ \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big).$$

- **Group-wise calibration** (Kleinberg et al., 2016):

$$\forall p \in [0,1],\ \ \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big) = p.$$

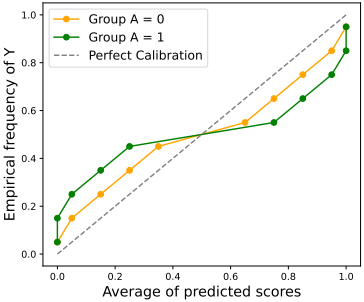
- **Multicalibration** (Hebert-Johnson et al., 2018): this notion is **stricter than group-wise calibration**, as it requires predictions to be calibrated for every subpopulation within a family of sets based on $\mathcal{X}$, where all sensitive groups are included in $\mathcal{X}$ (potentially overlapping groups).

Example: *Calibration parity*

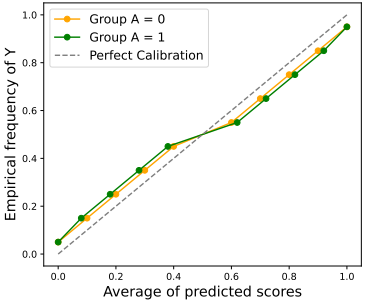
Consider a model s estimating the **probability that a policyholder experiences an accident** within a year. We evaluate the unfairness of s by requiring the **same calibration errors** for both sensitive groups $A = 0$ and $A = 1$:

$$\forall p \in [0,1], \;\; \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big)$$

Unfair model s



Fair model $\tilde{s}$

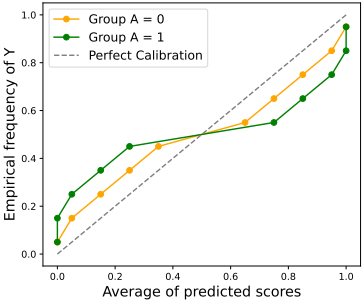


Example: *Group-wise calibration*

Consider a model s estimating the **probability of developing a disease** for different insureds. We evaluate the unfairness of s by requiring **no calibration errors** for both sensitive groups $A = 0$ and $A = 1$:

$$\forall p \in [0,1], \;\; \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 0\big) = \mathbb{P}\big(Y = 1 | s(\boldsymbol{X}, A) = p, A = 1\big) = p.$$

Unfair model s



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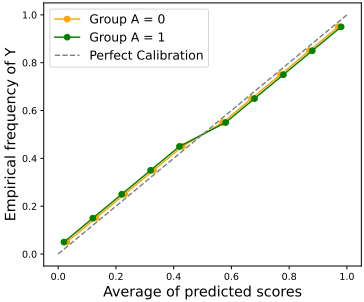

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Overview of Unfairness Mitigation Techniques

To correct for unfairness in model predictions, mitigation approaches can be applied at different stages of the machine learning pipeline:

1. **Pre-processing**: Modify training data to reduce bias **before model learning**.
2. **In-processing**: Incorporate fairness constraints **into the learning** algorithm.
3. **Post-processing**: Adjust model predictions **after model training**.

- In this study, we will consider two **post-processing mitigation approaches** to achieve **group-wise calibration** under the group fairness criterion of Sufficiency.

Post-processing Methods for Sufficiency

- Hebert-Johnson et al. (2018) propose an **iterative boosting algorithm** (the **HKRR** method): the algorithm works by iteratively identifying and removing calibration violations within subgroups until convergence.
- Haghtalab et al. (2023) applies game-theoretic concepts in the context of **multi-objective learning** (the **HJZ** method), where learning guarantees must hold simultaneously across a set of distributions and loss functions, defined here as calibration within each subgroup.

SEER dataset

- Prediction of **1-year mortality scores** for individuals diagnosed with skin cancer (`https://seer.cancer.gov`) in the United States (Sauce et al., 2023).
- The dataset contains $n = 547{,}878$ individuals from 2004 to 2018.
- We have the following explanatory variables, $(\boldsymbol{X}, A)$:
  - **Demographic data:** duration, age, gender, ethnicity, marital status.
  - **Medical data:** time since tumor onset, mitotic rate, primary tumor site, spread, tumor size, laterality, ulceration, metastases.
- The **sensitive attribute** considered here is **gender** (A) with two categories: **female** and **male**.

Group-wise Calibration Metric

- The calibration-based unfairness metric $\mathcal{U}$ of model s is calculated, based on the ECE, as

$$\mathcal{U}(s) := \max\left\{\mathrm{ECE}(s \mid A = \textbf{female}), \mathrm{ECE}(s \mid A = \textbf{male})\right\},$$

where the **ECE** for a model s, considering all observations, is defined as (Pakdaman Naeini et al., 2015)

$$\mathrm{ECE}(s) := \sum_{b \in [B]} \frac{n_b}{n} \cdot |\mathrm{freq}_b - \mathrm{conf}_b|$$

where the bins $1, \ldots, B$ are defined according to the quantiles of the predicted scores, and for n realizations $(\boldsymbol{x}_i, a_i, y_i)$ of $(\boldsymbol{X}, A, Y)$:

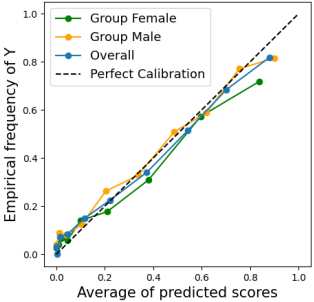
- $\mathrm{conf}_b = \frac{1}{n_b} \sum_{i \in b} s(\boldsymbol{x}_i, a_i)$,
- $\mathrm{freq}_b = \frac{1}{n_b} \sum_{i \in b} y_i$.

Application of Unfairness Mitigation Methods

We apply the two correction algorithms (HKRR and HJZ) on the SEER dataset.

- See the Appendix for more details on the procedure in practice.

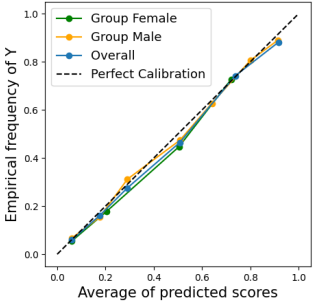
Initial model s



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.042

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.035

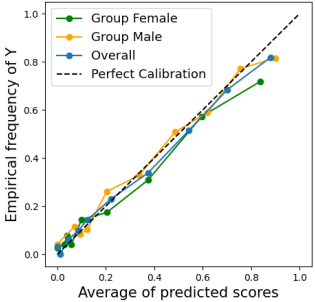
Fair model with HKRR



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.031

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.023

Fair model with HJZ



$\mathrm{ECE}(s \mid A = \textbf{female})$: 0.040

$\mathrm{ECE}(s \mid A = \textbf{male})$: 0.027

Future Work: Heterogeneity of Scores in Sensitive Groups

- Consider the case of a probabilistic classifier that estimates the probability of a "**positive outcome**" Y, such as credit approval or hospital admission.
- Due to historical injustices, confidence **score ranges may vary between sensitive groups**, with disadvantaged groups having lower scores.
- La Cava et al. (2023) introduced **Proportional Multicalibration** as a first step toward addressing this: "*By relating sufficiency to multicalibration (MC), we describe a shortcoming of MC that can occur when the outcome probabilities are strongly tied to group membership,* **MC does not provide equal protection to groups with different risk profiles**."

Conclusion

- **Calibration** is one type of metric used to assess the **performance of a binary probabilistic classifier**.
- When **evaluated across sensitive groups**, calibration can serve as an **unfairness metric**: it is widely used in healthcare and relevant in insurance.
- Future work: due to **variability in score ranges across sensitive groups**, mitigation algorithms may still favor majority groups.

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In Practice: Steps for the Mitigation Procedure

We can apply the **HKRR** and **HJZ** mitigation procedures using the approach, using the Python package `multicalibration` (Hansen et al., 2024):

1. Split the dataset into **three samples**: *train*, *calibration*, and *test*.
2. **Fit the predictive model** s on the *train* sample, using both $\boldsymbol{X}$ and A.
3. Apply the fitted model s on the *calibration* and *test* samples to predict the 1-year mortality scores.
4. **Fit the post-processing method**, denoted g_1 or g_2, on the *calibration* sample.
5. Apply the method g_1 or g_2 to obtain 1-year mortality scores calibrated by sensitive group on the *test* sample.

We will denote the two **fair predictors**: $\tilde{s}_1 = g_1 \circ s$ and $\tilde{s}_2 = g_2 \circ s$.